DAY - 10 08/01/2025

practice: HCF & LCM (word problems)

1) key Idea

HCF → used when we divide or group things equally.

LCM → used when events repeat together (time, bells, traffic lights).

2) Real word Ex.

Bells Tolling (LCM)

1) A bell rings every 6 min and another every 8 min. when will they ring together?

LCM of 6 and 8

$$\begin{array}{r|l} 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$3 \times 2^3 = 3 \times 8 \Rightarrow 24$

<u>Ans</u> They ring together after 24 min.

ii) Traffic Light (LCM)

One signal changes every 30 sec, another every 45 seconds. When will both change together?

2	30
3	15
5	5
	1

~~2~~	~~40~~
~~2~~	~~20~~
~~2~~	~~10~~
~~5~~	~~5~~
	~~1~~

5	45
3	9
3	3
	1

~~$2^3 \times 3 \times 5 \Rightarrow 8 \times 3 \times 5 \Rightarrow 2 \times 5 \Rightarrow$~~

$2 \times 3^2 \times 5 \Rightarrow 10 \times 9 \Rightarrow 90$ sec

iii) Distributing Item (HCF)

One signal changes every 30 seconds.
Find the largest number of students who can be divided equally among 48 pens and 60 pencils.

12	48
	4

12	60
	5

HCF (48, 60) = 12

Ans: 12 students

3) steps to solve word problem.

1) Read the Problem carefully

2) Id keywords

* together, again, repeat → LCM

* maximum, equal groups, divide → HCF

3) find HCF (or) LCM

4) write the final answer with units

(minutes, seconds, students).

6) practice Questions

1) Two bells ring at intervals of 12 min and 18 min. when will they ring together?

Lcm (12, 18)

~~6 | 12~~
~~2 | 2~~
~~1~~

~~9 | 18~~
~~2 | 2~~
~~1~~

2 | 12
2 | 6
3 | 3
1

2 | 18
3 | 9
3 | 3
1

~~2 × 6 × 9~~

$2^2 \times 3^2 \Rightarrow 4 \times 9 = 36$ min

The bells ring together at 36 min

2) find the HCF of 96 and 128

$$\begin{array}{r|l} 2 & 96 \\ \hline 2 & 48 \\ \hline 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 2 & 128 \\ \hline 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$96 = 2^5 \times 3$

$128 = 2^7$

$HCF = 2^5 \Rightarrow 32$

3) Three traffic lights flash at intervals of 10 sec, 15 sec, 20 sec. After how much time will all the light flash together?

$$\begin{array}{r|l} 2 & 10 \\ \hline 5 & 5 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 5 & 15 \\ \hline 3 & 3 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 2 & 20 \\ \hline 2 & 10 \\ \hline 5 & 5 \\ \hline & 1 \end{array}$$

$2^2 \times 3 \times 5 = 4 \times 3 \times 5 = 60$

Ans

All lights will flash together after ~~after~~ 60 seconds.